

Event

$E(t, x)$

A generative ambient instrument.

A single HTML file.

Never the same session twice.

"Time and space combined to generate a sonic experience."

Version 1.0

Yuwa / Positive Lofi · 2025-2026

I. Origin

It began with a patient suffering from tinnitus. A white noise audio file created to help them mask the ringing and sleep, posted on YouTube without much thought. Months passed. The file accumulated hundreds of thousands of hours of listening. Then millions.

This accidental gesture raised a question: why does white noise work so well? Because it is constant, predictable, surprising no one. The ear surrenders to it because it demands nothing. But is it possible to build something that holds the same promise while carrying a genuine musical intention? Something that breathes, evolves, never repeats?

Event is the answer to that question. Not an improvement on white noise – a break from it. The instrument keeps the promise of a continuous sonic background, but replaces it with a living harmonic space, an endless composition, a sonic organism.

In cetacean anatomy, the blowhole is the organ through which they breathe and vocalize. Humpback whales have songs that drift culturally – an entire population can change its sonic repertoire within weeks, adopting a new song propagated across the ocean. Over a decade, the initial and final versions are unrecognizable from one another, yet the transition was continuous, gentle, never brutal. Event works the same way: it drifts harmonically, tonally, timbrally – but always within musical coherence. Event breathes.

II. The Postulate – Entropic Seed and Universal Time

At the heart of Event lies a radical philosophical claim: two sessions cannot be identical because they cannot begin at the same place in the universe.

random.org – cosmic noise

When Event starts, it contacts random.org before producing a single sound. This service generates true randomness from atmospheric radio noise – the natural electromagnetic fluctuations of the Earth's atmosphere, captured by physical antennas. This is not a pseudo-random algorithm. It is real entropy, drawn from the physical world.

This radio noise results from thousands of simultaneous phenomena: storms somewhere on Earth, cosmic radiation, solar activity, human and natural electromagnetic emissions. At every instant, it is unique. It will never be reproduced exactly.

Event draws a 32-bit integer from random.org – eight hexadecimal characters, displayed in the control panel. This is the session seed. Everything that follows – tonalities, weightings, timings, transitions – will flow from this initial value.

The precision timestamp

The seed alone is not enough. Event combines the entropic seed with two timestamps:

Date.now() – Unix time in milliseconds since January 1st, 1970 00:00:00 UTC. This number is universal: the same value everywhere on Earth at the same instant, regardless of time zones.

performance.now() – time elapsed since the page loaded, in microseconds. This number is local, specific to this browser instance, to this particular session.

The combination of these three elements – atmospheric entropy, universal time, local time – produces a seed that will exist only once in the history of the universe. Not as a metaphor. Because it is mathematically true.

The seed as identity

The seed is displayed in hexadecimal in the panel – eight characters, like **A3F2C1B8**. It allows you to return to the same starting point, but not to reproduce what happened next. The same seed in a different environment will yield an identical beginning but a session that diverges progressively. That is the distinction between seed and session. The seed is reproducible. The session is ephemeral.

III. Composition by System

Event is not music generated by artificial intelligence. This distinction matters – it touches on authorship, intention, and what it means to compose.

A generative AI system learns from an existing corpus and produces outputs that statistically resemble that corpus. It has no rules of its own – it has patterns extracted from other music. Event works differently. All its rules are written explicitly: which modes, which preferred intervals, which stability for which harmonic degrees, which density for which dynamic state. These rules belong to their author. The system plays them – it does not invent them.

This practice is ancient. Xenakis used stochastic formulas to compose *Achorripsis* in 1957. Brian Eno, with *Music for Airports* (1978), laid down composition rules for musicians to execute. Steve Reich explored phase patterns – simple rules producing complex, unpredictable results. Éliane Radigue meticulously calibrated interactions between analog oscillators, letting the instrument find its own beating.

Event belongs to this tradition. It is an algorithmic score – a set of musical rules defining a space of possibilities, which the system explores in real time. The author composed the patch. The instrument plays.

"It's a modular synth session whose patch I composed."

Ephemerality as value

The decision not to auto-save sessions is not a technical limitation. It is a philosophical choice. An Event session is an event in the physical sense – it exists in time, it occupies a moment, it will not recur. The seed exists precisely for cases where you want to return to a starting point. But it does not save the session: it opens a door to the same corridor without guaranteeing you will find the same rooms.

IV. The Botanica Identity

Botanica is not a preset among others. It is the instrument's DNA – the sonic philosophy that informs every technical decision, every weighting, every compromise.

A forest is never silent. Even in the deepest calm, something is always happening: a leaf moving, a wood creak, a distant insect, the wind through branches. These events are not planned. They emerge from the physics of the place. And yet they form a coherent whole. You recognize a forest by its sound.

That is exactly what Event seeks to create. Not music in the sense of sounds organized in time. But an inhabited sonic space – a texture with a recognizable identity, evolving organically, never saturating.

Botanica density

Botanica density follows a simple rule: something is always happening, but nothing overwhelms anything else. Constant micro-events rather than large sonic masses. Pads exist but are short. The drone is always present but its gain is so low you don't hear it as a drone – you sense it as coherence. The dominant dynamic state is flowing – neither the silence of sparse nor the saturation of active.

Inhabited silence

In sparse state, Event does not go silent – it lightens. Foley continues (a drop, a creak, leaves), micro-motifs appear every 3 to 8 seconds, organic percussion maintains a light presence. What you hear as silence is not: it is the background noise of the space, proof that the instrument is alive even when it is not playing.

The subliminal drone

Event's drone is designed to never be a target. Five sine oscillators whose frequencies are modulated by LFOs at irrational ratios (ϕ , π , e). These irrational ratios prevent regular beating – frequencies never meet periodically, so there is no audible pulse. The result is a presence that is felt without being heard. When the tonality changes, the drone silently glides toward the new fundamental over several minutes – inaudible as a transition, coherent as a result.

V. Harmonic System

The tonality pool

Event has 102 available tonalities – 11 modes across 12 roots, weighted by Botanica affinity.

Mode	Weight	Emotional character
Penta m / Penta M	4.0	universal, contemplative, no semitones
Dorian	3.5	dark jazz, softly melancholic, major 6th
Lydian	3.0	floating, unreal, augmented 4th
Japanese	2.5	minimalist, In pentatonic, widely spaced
Mixolydian	2.0	warm, bluesy, minor 7th
Major	1.2	luminous, resolved – contrast
Minor	1.0	melancholic, deep – contrast
Phrygian	0.6	mysterious, dark – rare
Lyd. augmented	0.5	unreal, Messiaen – rare
Locrian	0.3	tense, unstable – very rare

Root notes are also weighted: Ab, Eb, F, C and Bb are slightly favored (warmer, flat-key colors), while sharp roots (C#, F#, B) are less probable. This asymmetry is subtle – all roots appear – but it gives the instrument a slightly flat-key overall color.

Pivot harmonic transitions

A harmonic pivot is a note that belongs simultaneously to two scales. It is the bridge between them. Event calculates available pivots between the current tonality and each potential target. Stability score for each common note:

Degree	Stability	Example (C Major)
I – tonic	3.0	C
V – fifth	2.5	G
III – third	2.0	E
II / IV / VI	1.0	D / F / A
VII	0.5	B

Pivot score = stability in the departure scale × stability in the target scale. Example: the fifth of Ab Lydian (Eb) is the tonic of Eb Dorian – score $2.5 \times 3.0 = 7.5$. The ideal pivot between these two scales.

The transition unfolds in 3 phases: **Phase 1** (3–6s) – convergence toward the pivot. **Phase 2** (4–8s) – holding the pivot, both scales coexist, no clash possible. **Phase 3** (4–7s) – opening toward the new tonic. Total duration: 11 to 21 seconds.

VI. The Dramatic Arc

Event has a memory of itself at the scale of the session. An intensity curve modulates probabilistic biases according to session age – it forces nothing, it weights the dice.

The psychology of attention

We live in the TikTok era. An ambient piece beginning with 20 seconds of introductory silence will lose its listener before it even exists. Event is designed to work in this context without surrendering its nature. The solution is not to become less ambient. It is to be dense immediately, then allow breathing once attention is captured.

Segment	Duration	Character
Hook	0-15s	Dense immediately. First note at 3-5s.
Installation	15s-2min	The space establishes itself.
Breathing	2-4min	Dip. Slow harmonic transitions.
Rise	4-7min	Progressive return. Richer than before.
Climax	7-9min	Maximum density. Solo possible.
Dissolution	9-11min	Progressive withdrawal.
Free flow	11min+	Pure probabilistic. No end.

The arc does not compose – it weights. At 7 minutes, Event does not necessarily play a climax. It makes climax more probable. A long silence can appear at the densest moment if the dice fall that way. The arc is an intention, not a score.

VII. Floating Instrument Weights

Event has abandoned the concept of rigid orchestrations – predefined palettes that disabled certain instruments for minutes at a time. Instead: floating weights. Each instrument always has a probability of triggering – never zero, never fixed. These probabilities drift slowly through a blend mechanism between 8 sonic colors.

Each color boosts certain instruments above the baseline. These colors blend continuously over several minutes. The ear does not perceive the moment of change – it perceives a drift in texture, like lighting that shifts imperceptibly.

Every instrument has a baseline floor of 0.25 minimum – prepared piano, organ, voice, metal are always potentially present from the first minute.

Solos

Every 2.5 to 5 minutes, Event triggers a solo – a melodic fragment of 4 to 7 notes played by the instrument with the highest floating weight at that moment. It is the most 'composed' instant of the session: a clear melodic line emerging from the ambient flow.

VIII. User Guide

Starting

Open `event_v1.html` in Chrome. Click anywhere to wake the sound. The first note falls within 3 to 5 seconds. A second click pauses with a 4-second fade.

Control panel

Click `...` at the bottom of the screen.

Performance modes

`normal` – dramatic arc active, free flow.

`experimental` – long silences assumed, glitches more present.

`concert` – permanent density, very brief silences.

`original` ■ – reset to Ab Major, initial state.

Recording for DAW

1. Lock scale + root (e.g. D Lydian for something floating and luminous).
2. Launch ● wav from the panel.
3. Let it run. Everything produced is harmonically coherent in that space.
4. Cut interesting passages in your DAW.

WAV export: 32-bit float stereo. Video export: 1080p VP9.

Stream

In OBS: add a Browser Capture source. Enable audio capture. Event produces continuous video and audio, never identical.

Tonal box

Enable 'backing' in the panel. During a harmonic transition, it displays the pivot bridge: departing scale fading, pivot note ■ at center, new scale rising.

Seed

Note the seed to return to a starting point. The session restarts from that point – but what happened between two moments will never recur.

IX. Technical Notes

Browsers

Chrome 66+ recommended. Firefox 76+, Safari 14.1+, Edge 79+ supported.

Audio chain

All instruments pass through a global bus (instrBus) before the master. Chain: instrBus → musical compressor (−12dB, ratio 2:1, release 800ms) → high-shelf −4dB at 3kHz → limiter (−6dB, 20:1) → true peak (−1dB, 20:1) → output.

Single file

One standalone HTML file (~200KB). No external dependencies except Google Fonts (DM Mono, Cormorant). No server, no framework, no installation. Event must be shareable by email, postable on GitHub, openable from Files. The friction between 'having the file' and 'hearing the sound' must be zero.

No AI

Event contains no trained model, no music dataset, no neural network. Its rules are hand-written composition logic – explicit probabilities, weighted scales, harmonic constraints defined note by note. What the instrument does is determined by its author's decisions, not by statistical patterns extracted from other music.

License

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